SHAHSAWAR ORAKZAI

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PROFESSIONAL SUMMARY

AI Engineer specializing in production **RAG architectures**, autonomous **multi-agent systems (LangGraph/LangChain)**, and high-concurrency **FastAPI** backends. Track record of architecting and shipping resilient systems end-to-end: **sub-250ms semantic retrieval** across **10,000+ document chunks**, multi-step agentic workflows with persistent state, and data pipelines processing **20k+ daily records**. Experienced technical owner operating autonomously for international teams in fast-paced, async environments.

EXPERIENCE

Freelancer — Feb 2026 – Present
AI Engineer (Contract) — Remote

- Delivered enterprise **RAG assistants** and document retrieval systems for business clients with **FastAPI/pgvector** backends and responsive **TypeScript** frontends, cutting manual inquiry response times by **~65%**.
- Orchestrated autonomous **agentic workflows** using **LangChain**, open-source LLMs, and function-calling architectures with persistent memory and external tool integration, eliminating **manual intervention** across multi-step operational processes for multiple clients.

MindGigs — Dec 2025 – Feb 2026
Python Engineer — Hybrid

- Engineered high-throughput **REST APIs** using **FastAPI** and **asyncio**, boosting request throughput by **~40%** with strict **Pydantic** schema validation.
- Integrated locally-hosted **open-source LLMs** and developed **Dockerized** data scraping pipelines processing **20k+ daily records**.

Pakistan Swimming Federation — May 2024 – Sep 2025
Data and Automation Engineer — On-site

- Designed and deployed a **Python**-based MIS for **200+ registered athletes**, serving as sole technical owner and replacing paper-based records with a searchable relational database.
- Automated administrative workflows using **Python** and relational schemas, eliminating manual data entry for **30+ provincial swimmers** across district and national championships.

PROJECTS

BOTRON (RAG Chatbot & 3D Web App) — Three.js, WebGL, FastAPI, LLM APIs, Supabase, GitHub Actions
Source Code

- Architected an interactive, scroll-driven **WebGL** application featuring 3D **inverse-kinematics** gaze tracking and adaptive frame rates (60/30/0 FPS), deployed on **GitHub Pages**. Pre-launch content reached **50,000+ impressions** and **500+ likes** on LinkedIn.
- Deployed **BOTRON**, a production **RAG** chatbot achieving **95%+ retrieval accuracy** across a **100+ question evaluation dataset**, with **FastEmbed** semantic search, similarity-floor hallucination prevention, and 3-model **LLM** failover with cloud-hosted backend.
- Integrated **Supabase** for real-time telemetry and query logging, tracking user interaction patterns to continuously optimize prompt relevance and system latency.

Rednote (Document Retrieval Assistant) — Python, pgvector, Neon PostgreSQL, TypeScript, Docker, CI/CD
Source Code

- Developed a full-stack document retrieval assistant with **LLM-powered search** and **pgvector** semantic indexing, achieving **sub-250ms query response times** across **10,000+ indexed chunks**.
- Engineered an automated document ingestion pipeline with **JWT** authentication and guest-to-registered conversion, onboarding **30+ active users**.
- Containerized application with **Docker** and deployed to production via automated **GitHub Actions CI/CD** pipeline, reducing deployment cycles to under 3 minutes.

Async Extraction & ETL Engine — Python, curl_cffi, Neon PostgreSQL, TypeScript, LLM APIs, Docker
Source Code

- Engineered an async extraction engine using **curl_cffi** with browser TLS fingerprint impersonation to navigate Cloudflare bot management defenses, concurrently scraping **30+ web pages in parallel**.
- Architected an **8+ LLM API rotation pool** with adaptive rate-limiting and automatic failover, structuring raw HTML into schema-validated **JSONL** datasets for downstream AI and data pipelines.
- Deployed on a serverless cloud stack backed by a **Neon PostgreSQL** database, featuring a **TypeScript** UI with real-time polling that cut manual verification overhead by **90%**.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, C++, SQL (PostgreSQL), Bash
AI / LLM	RAG Architectures, AI Agents, LangChain, LangGraph, n8n AI Workflows, LangSmith, pgvector, FastEmbed, OpenAI, Anthropic APIs, Open-Source LLMs, ETL Pipelines
Backend	FastAPI, Pydantic, SQLAlchemy, AsyncIO, Supabase, Neon PostgreSQL, REST APIs, WebSockets
Frontend	Three.js, WebGL, GSAP, HTML5, CSS3, React, Vite
DevOps & Tools	Docker, Git, GitHub Actions, CI/CD, Cloud Platforms, Linux, Hugging Face

EDUCATION

University of Agriculture, Peshawar — Expected 2028
Bachelor of Science in Computer Science (BSCS) — CGPA: 4.0

CERTIFICATIONS & ACHIEVEMENTS

CS50X: Computer Science (Harvard University) Verify — Verified Credential
CS50P: Programming with Python (Harvard University) Verify — Verified Credential

National Athlete: Represented provincial swim team for 10 years as a National Swimmer, winning 10+ medals including Gold at National Junior Championships.